2 SOFTWARE

The programs that run on a computer are called software. Programs are instructions that are carried out or executed by the computer's processor, which provides other instructions for the rest of the computer.

Some computers can only run one program at a time, while others can run multiple programs at the same time. Some computers are single purpose, which means that they are only able to run one piece of software (for example, the software used to control a microwave). Other computers are general purpose, which means that they are able to carry out a range of tasks (for example, the software used in a tablet computer).

There are two basic groups of software: system software and application software.



LEARNING OBJECTIVES

- Be able to identify the purpose of systems software and applications software
- Know about operating systems and system software tools such as utilities
- Know about the role/function of the operating system, including basic knowledge of:
  - single user and network
  - memory management
  - resource management
  - security
  - print spooling
- Know about software applications (apps), including office-productivity tools, web authoring, image and sound editing, presentation software, control software, project management software
- Know about software licensing types (free/open source and proprietary software)
- Understand that the purpose of communication software is to provide remote access to systems and to exchange files and messages in text, images, audio and/or video formats between different computers or users
- Know why software is updated, how it is done and possible risks to data/systems
- Understand that settings of ICT systems can be configured to meet the accessibility needs of individuals
- Be able to justify choices made in identifying and configuring hardware and software

SYSTEM SOFTWARE

Programs that are designed to maintain or operate the computer system are known as system software.

UTILITY SOFTWARE

Utility software is one form of system software, which carries out configuration and maintenance tasks.

SUBJECT VOCABULARY

utility software (also known as utilities) system software that carries out configuration and maintenance tasks

BACKUP

Backup utilities create a copy of files and programs. Backups can be set to run automatically (usually at a time when the system is not in use) or can be started by a user.

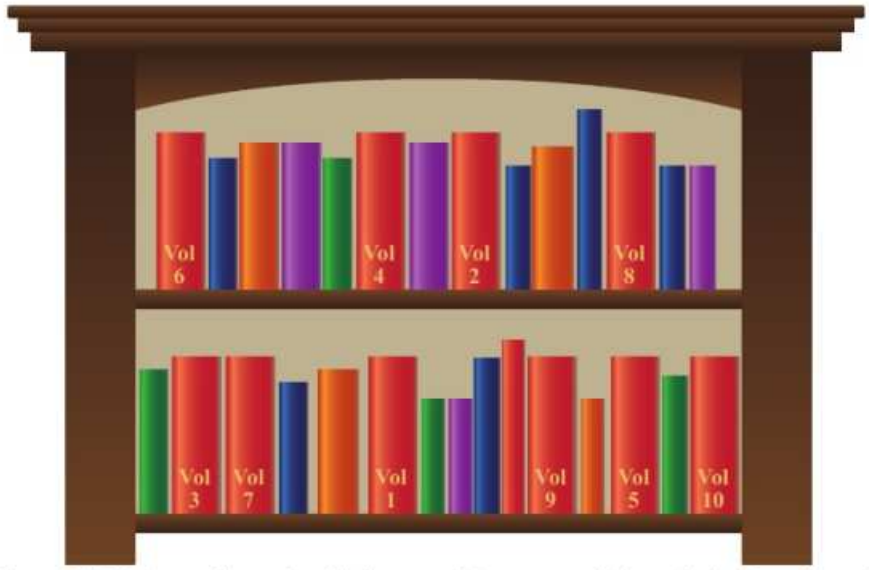
You will learn more about backups in *Unit 3 Operating online* (pages 104–105).

DEFRAGMENTATION

GENERAL VOCABULARY

fragmented broken into pieces

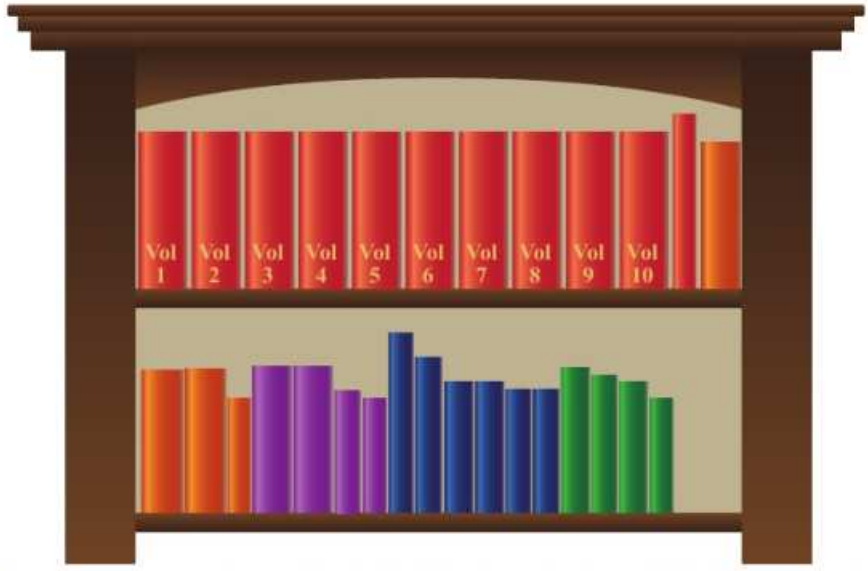
As data is stored to hard disk, some systems spread it across the disk wherever there is free space. This means that sometimes data is **fragmented** and stored out of order or out of sequence.



▲ **Figure 2.1** Data is sometimes stored in the nearest free space, which results in data getting out of sequence. This is an illustrative example.

Data from the same file may be spread over multiple locations on a platter, on different platters in the cylinder or sometimes even on different disks. When these files are loaded, data has to be retrieved from these multiple locations. The read/write head of the hard drive has to find this data, which takes extra time and slows down the process of loading the files.

To avoid this and to speed up the system, disk defragmentation utilities reorder the fragments of data so that they are stored as close to each other as possible.



▲ **Figure 2.2** Defragmentation puts the data back in order so that the read/write head does not have to search the disk for fragments of data. This is an illustrative example.

Defragmentation utilities are usually set to run automatically, but they can also be started by a user.

SKILLS EXECUTIVE FUNCTION
INTELLECTUAL INTEREST AND CURIOSITY
SELF-DIRECTION

ACTIVITY

▼ DEFRAGMENTATION

1 Run a defragmenter analysis on your system. When was the analysis last run? How fragmented was the data last time the analysis ran?

2 Research the need for defragmentation on SSDs.

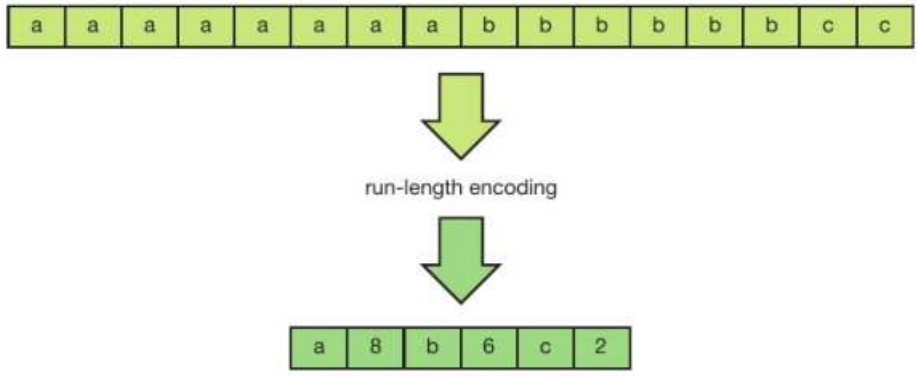
COMPRESSION



▲ **Figure 2.3** Data can be compressed to remove unnecessary use of code or space in the file. This is an illustrative example.

Compression utilities reduce the original size of a file or set of files.

Where there are repeated patterns of data, rather than storing every repeated instance, only the first instance of the data is stored, alongside how many times it is repeated. For example, in Figure 2.4, there were eight instances of a, six instances of b and two instances of c in the original file. This data can be compressed to be stored as 'a8b6c2', which reduces the amount of storage needed.



▲ **Figure 2.4** Run Length Encoding is one method of data compression

SUBJECT VOCABULARY

decompress extract data from a compressed file so that it can be read

The resulting compressed file is not usually readable by the original application. Compression utilities also have the ability to **decompress** (extract) the data from a compressed file so it can be read by the original application again.

FORMATTING

Disk formatting prepares storage media such as a hard disk drive or USB flash drive for its first use. If a disk has already been used, then formatting it will make all of the data on the disk unreadable by normal applications.

DID YOU KNOW?

Different operating systems require disks to be formatted in certain ways in order to perform specialist tasks. Disks can also be formatted in a way that makes them compatible with all operating systems.

Disk formatting utilities make this process easy for users by providing them with the option to choose the file system, file unit sizes and the name of the disk. Perhaps most importantly, they make sure that the user really wants to format the disk.



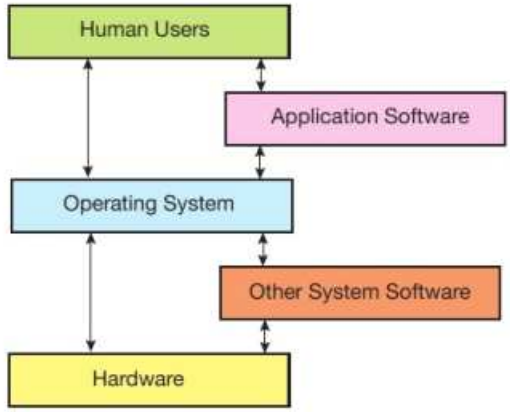
▲ **Figure 2.5** Accidentally formatting the wrong disk could result in the unwanted loss of data

DID YOU KNOW?

Formatting a disk will not erase the data completely and it can still be recovered using specialist tools, but there are some utilities which will overwrite the data so that files cannot be recovered.

OPERATING SYSTEMS

The operating system (OS) allows the user to control and manage the computer's hardware.



▲ **Figure 2.6** The operating system lets the user control the hardware, either directly or by using applications software or system software

GENERAL VOCABULARY

customise to change something in order to make it more suitable for a particular user or to make it look special or different from things of a similar type

Single-user operating systems only allow for a general user and do not provide the option to **customise** the user interface for different users. They are often found in household appliances.

Network operating systems have additional functionality, including:

- sending requests to a server when users log in with their username and password
- separating user accounts and ensuring that users cannot access each other's files
- providing access to network storage and shared resources such as networked printers.

MEMORY MANAGEMENT

The operating system allocates the required amount of memory (RAM) to one or more applications. When the application no longer requires the memory space, the operating system makes the space available for other applications to use.

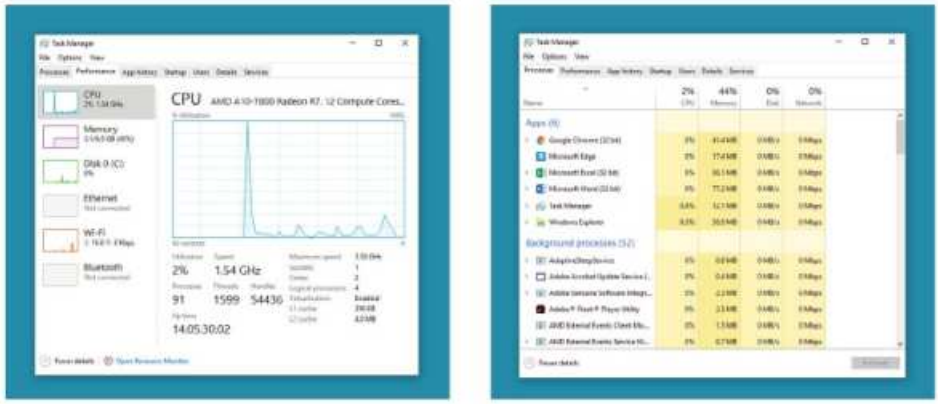
The operating system also handles the creation of virtual memory. See page 60 for more information about virtual memory.

RESOURCE MANAGEMENT

SUBJECT VOCABULARY

graphics card a device built into or added into a computer to allow it to display visual graphics

System resources include internal components, such as the processor and **graphics card**, and external devices, such as printers. As well as allocating memory, the operating system ensures that system resources are made available to applications when they are required. If a system resource is already being used by another application, then the operating system may say that the required resource is in use and place the additional request in a queue. For some resources, such as printers, the operating system can tell the user when the resource becomes available.



▲ **Figure 2.7** Information about the use of memory and resources is provided by utility software

SECURITY

SUBJECT VOCABULARY

authentication the process of confirming that a user is permitted to access certain files, hardware and software

firewall a system (hardware or software) that protects a computer network from being used or looked at by people who do not have permission to do so

Operating systems can give users secure access to a computer's storage and other hardware through usernames and passwords, biometric scanning (see pages 27–28) or personal access cards. This process is called **authentication**. Operating systems can provide software **firewalls** to authorise or prevent network data from remotely accessing a service or application.

PRINT SPOOLING

SUBJECT VOCABULARY

print spooling the process of keeping pages queued in order, ready to be printed by a printer

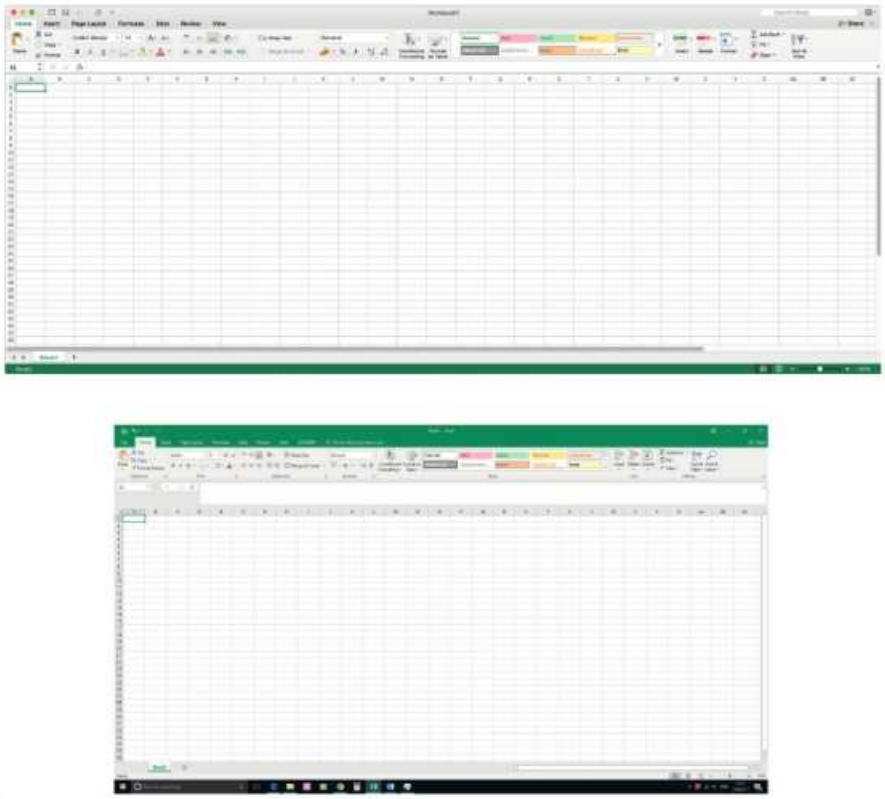
During large print jobs, the computer will have the pages ready for the printer faster than the printer can produce them. The operating system keeps each page in a queue ready for printing. This process is called **print spooling**.

APPLICATION SOFTWARE

Software applications (apps) allow users to produce a digital product, such as a presentation or image, or carry out specific tasks that are not related to the operating system.

Apps are usually downloaded from servers on the internet. Some devices, like smartphones and tablets, use operating systems that provide access to app stores. Some of these only make apps available to users after the store has checked the quality of the app and has ensured that the app does not include code that will harm a user's device or data. Apps can also be installed from storage media like DVDs.

You will learn more about online software in *Unit 4 Online goods and services* (178–179).



▲ **Figure 2.8** Some developers provide versions of their apps for different types of computer and operating systems. The top window shows the Apple macOS version whilst the bottom window shows the Microsoft® Windows version.

HINT

When you are writing about application software in your exam, you should refer to the application by its type rather than by a particular name. For example, use the term 'word processing software' rather than the brand name Microsoft® Word.

You need to be able to identify the type of software that meets specific needs. Each type of application software has particular features that make it suitable to perform a particular function. Sometimes more than one type of application can complete the task, but you should consider which one is most suitable.

OFFICE PRODUCTIVITY SOFTWARE

SUBJECT VOCABULARY

application suite a collection of application software that share the same look and user interface; often, they can share data between each other and share some functionality

hyperlink a link that can be clicked in order to go to another location (often a web page on the internet)

Office productivity software is any application that can carry out work-related tasks. They are often available as an **application suite**.

WORD PROCESSING

Word processing software lets users create documents that mostly include words, but also some images. They can also include tables, **hyperlinks**, equations, simple drawings and shapes and charts. An example of word processing software is Writer from the LibreOffice application suite.

Word processors are a good choice of application for creating:

- letters
- reports
- essays
- books (for example, this book was written using word processing software).

Some word processing applications allow people to work together on a document. Comments can be added to the text so that others can understand the writer's intention and give feedback. The software can also track any changes made to the document, giving others the option to accept or reject those changes.

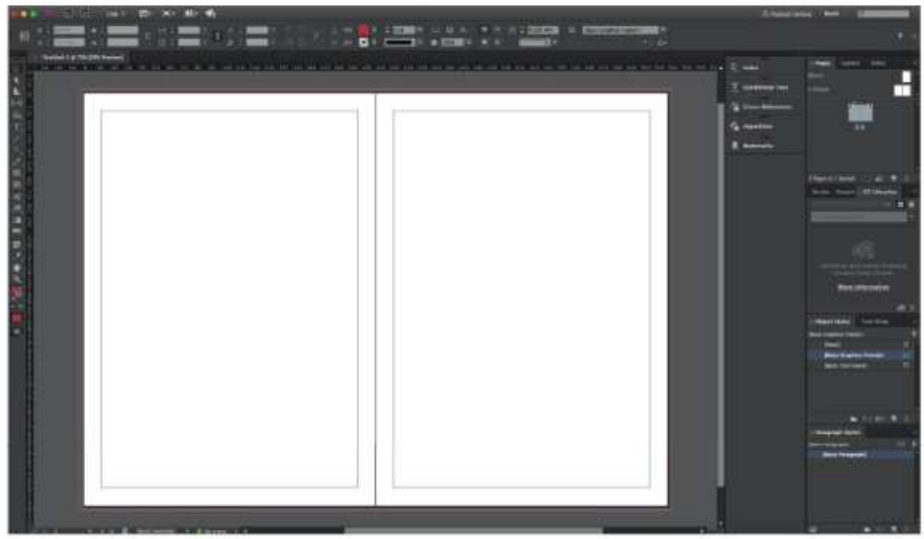
SUBJECT VOCABULARY

font the size and style of text

Word processing applications allow users to change the **font** and provide different layout views to allow users to position text and images on the page. They usually come with spelling and grammar checking tools that identify errors in the text as the user types.

DESKTOP PUBLISHING (DTP)

DTP software has many similar features to those in word processing software. The main difference is the way in which the software allows users to work with complicated page layouts. An example of DTP software is Adobe® InDesign®[2].



▲ **Figure 2.9** Page layouts in DTP software help the designer to see what the final version will look like when printed

SPREADSHEET

Spreadsheet applications are used to do calculations. Users can use and create formulae and functions to perform automatic calculations on values that can be entered and changed later. These features allow spreadsheets to be used to model financial scenarios and answer 'what if?' questions, such as '**What** would the cost per ticket be **if** I changed the maximum number of staff required at my event?'.

Venue	Gold	1		3000
Catering package	Gold	1		30
Entertainment	Bronze	3		300
Decorations	Silver	2		300
Transport	Gold	1		80
Number of Staff required				20
Ticket price needed			£	59.25
Meets requirements?				No

Venue	Gold	1		3000
Catering package	Gold	1		30
Entertainment	Bronze	3		300
Decorations	Silver	2		300
Transport	Gold	1		80
Number of Staff required				30
Ticket price needed			£	55.92
Meets requirements?				No

▲ **Figure 2.10** Two scenarios from a financial model

Venue	Gold	1	=VLOOKUP(C11,$L$3:$R$5,5)
Catering package	Gold	1	=VLOOKUP(C12,$L$3:$R$5,3)
Entertainment	Bronze	3	=VLOOKUP(C13,$L$3:$R$5,4)
Decorations	Silver	2	=VLOOKUP(C14,$L$3:$R$5,6)
Transport	Gold	1	=VLOOKUP(C15,$L$3:$R$5,7)
Number of Staff required			=ROUNDUP(D5/10,0)
Ticket price needed			=(G14-J14)/D5
Meets requirements?			=IF(D19>D8,"No","Yes")

▲ **Figure 2.11** Formulae and functions are used for calculations that are carried out automatically when values change

2 ADOBE AND ADOBE INDESIGN ARE EITHER REGISTERED TRADEMARKS OR TRADEMARKS OF ADOBE SYSTEMS INCORPORATED IN THE UNITED STATES AND/OR OTHER COUNTRIES

An example of spreadsheet software is Microsoft® Excel®.

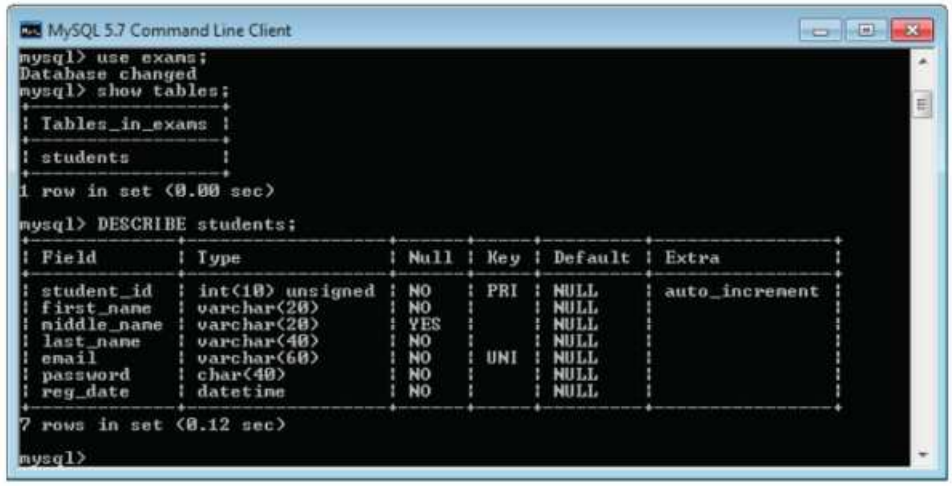
You will learn more about spreadsheets in *Unit 5 Applying Information and Communication Technology* (page 185) and *Unit 6 Software skills* (page 276).

DATABASE

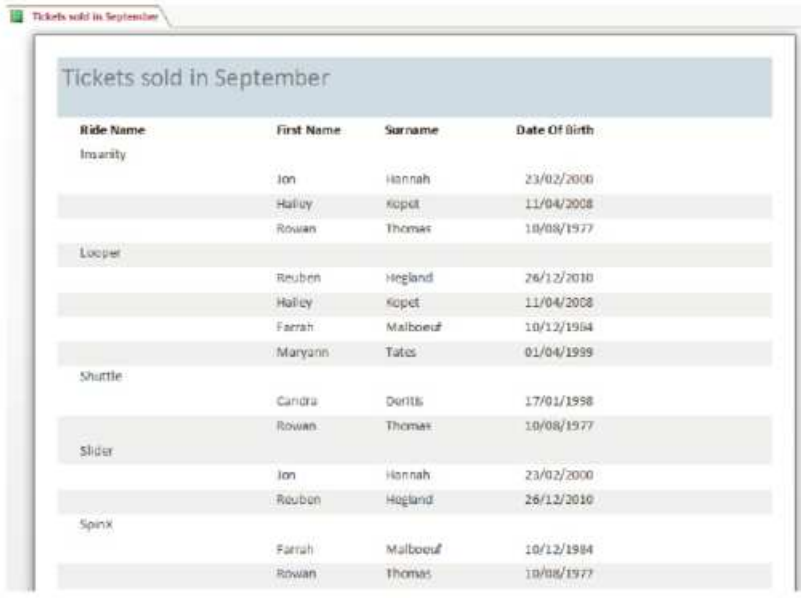
Database management systems (DBMS) are used to enter, edit and search data. Some systems can also produce reports that dynamically display real-time changes to the data as it updates.

DBMS software features include:

- data entry forms so that users can input data
- query editors so that users can select all data that meets certain criteria
- report builders so that users can display data in a more readable format.



▲ **Figure 2.12** This DBMS software uses a command line interface and contains fewer features



▲ **Figure 2.13** A report from a database

An example of DBMS software is Oracle® MySQL.

You will learn more about databases in *Unit 6 Software skills* (pages 307–333).

WEB AUTHORING

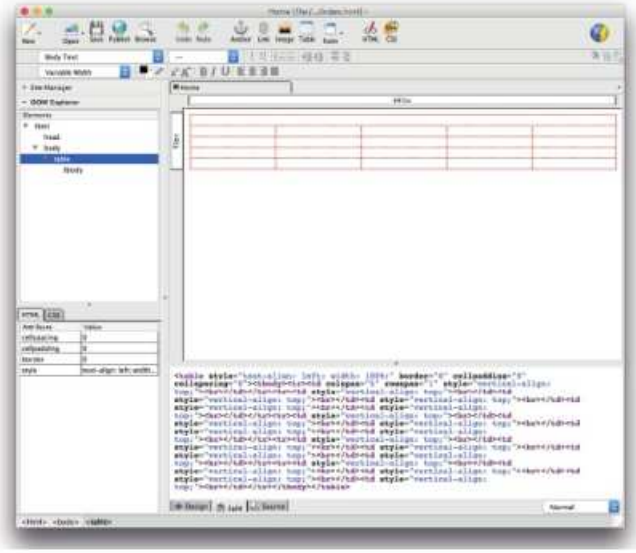
SUBJECT VOCABULARY

Hypertext Markup Language (HTML) a computing language read by web browser software
intranet a computer network used for exchanging or seeing information within a company
web server a computer that stores web pages and sends them to other devices that request them (often using web browser software)

Web authoring software lets users create web pages that include text and images. The pages are output as **Hypertext Markup Language** (HTML). HTML is read by web browser software such as Google® Chrome™[3], Microsoft® Edge or Mozilla® Firefox®[4]. Web browsers translate HTML into pages that people can see and read. Web pages can be linked together to create a website for people to view on an **intranet** or on a **web server** on the internet.

You will learn more about web browsers on page 55 and in *Unit 2 Connectivity* (page 87).

Although some people prefer to create or customise web pages by writing their own HTML, some applications will allow users to produce complex websites with little or no experience of HTML coding. These applications are sometimes called 'What You See Is What You Get' (WYSIWYG) editors, because the published page will look very similar to the page that the user edits in the web authoring application. An example of web authoring software is Adobe® Dreamweaver®[5].



▲ **Figure 2.14** Some web authoring software can display the code and design views one after the other

IMAGE EDITING

SUBJECT VOCABULARY

bitmap a computer image that is stored or printed as an arrangement of bits
vector graphic a graphical image made up of points and lines

Image editing applications let users create and change **bitmap** images, such as digital photographs, or **vector graphics**, such as drawings or logos.

GRAPHICS EDITORS

Graphics editing applications allow users to create or edit vector graphics. An example of graphics editing software is Serif® Affinity Designer[6]. Features of graphics editors include:

- vectorising images (that is, converting or tracing bitmap graphics to vector graphics)
- layers (that is, having the ability to place some graphics on top of others)
- adding text
- adding or drawing shapes and lines
- resizing, aligning or moving shapes and lines
- altering the colour of shapes, lines and fill areas.

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4 MOZILLA FIREFOX IS A TRADEMARK OF THE MOZILLA FOUNDATION
5 ADOBE AND ADOBE INDESIGN ARE EITHER REGISTERED TRADEMARKS OR TRADEMARKS OF ADOBE SYSTEMS INCORPORATED IN THE UNITED STATES AND/OR OTHER COUNTRIES
6 "SERIF" AND "AFFINITY" ARE BOTH REGISTERED TRADEMARKS OF SERIF EUROPE LTD

PHOTO EDITORS

These allow users to edit and enhance digital photographs and images. An example of photo editing software is GNU Image Manipulation Program (GIMP). Features of photo editors include:

- adjusting brightness and contrast
- resizing the image
- altering sharpness and blurring
- applying filters and effects such as distortion
- red-eye removal
- **cropping**.

SUBJECT VOCABULARY

cropping using a photo editor to remove the edges of an image

SKILLS INTELLECTUAL INTEREST AND CURIOSITY
ETHICS
COMMUNICATION
INTERPERSONAL SKILLS

ACTIVITY

▼ EDITING IMAGES

Go to the website of the cosmetics company, Dove, and search for the key terms 'evolution video' to find the video *Evolution*, which is about the way in which the beauty industry uses image editing applications. Watch the video, then discuss the impact that image editing applications can have on society.



▲ **Figure 2.15** There are many photo editing apps available for smartphones; finished images can be stored to local storage or shared online

SOUND EDITING

Sound editing software allows users to edit audio files or to join together different audio files in order to create multitrack music or soundtracks for video. An example of sound editing software is Audacity®. Features of this software include:

SUBJECT VOCABULARY

mute silence an audio track
solo only play one particular audio track
tempo the speed at which a track is played

- cut and join audio clips
- **mute** and **solo** some audio tracks
- alter volume levels for individual tracks
- change **tempo**

GENERAL VOCABULARY

reverberation prolonging a sound

SUBJECT VOCABULARY

normalisation increasing the average volume of a piece of audio to a defined maximum level

- frequency equalisation (changing the levels of high and low pitch frequencies)
- add effects like **reverberation**
- apply audio processing:
  - reverse
  - noise reduction
  - **normalisation**
  - fade the volume in, so that it gets louder, or out, so that it gets quieter.

PRESENTATION SOFTWARE

SUBJECT VOCABULARY

slide a single page of a presentation
pathway the sequence of or route through a series of slides

Presentation software allows users to create engaging multimedia content, including images, text, animation and video. This content can be placed on **slides** or **pathways** that are used to illustrate and support the spoken content of a talk given to an audience. An example of presentation software is Prezi™[7].

Some applications allow users to practise timings and add narration that automatically plays back on the appropriate slides. These slides can be set to advance automatically after a set period of time, or on command from the presenter. Notes can be added to the presentation to remind and prompt the presenter to speak about specific points during the presentation.

CONTROL APPLICATIONS

Control applications are used to make something happen in the physical environment. This type of software is often used to automate the movement of control devices or actuators, such as motors. See page 37 for more information about actuators.

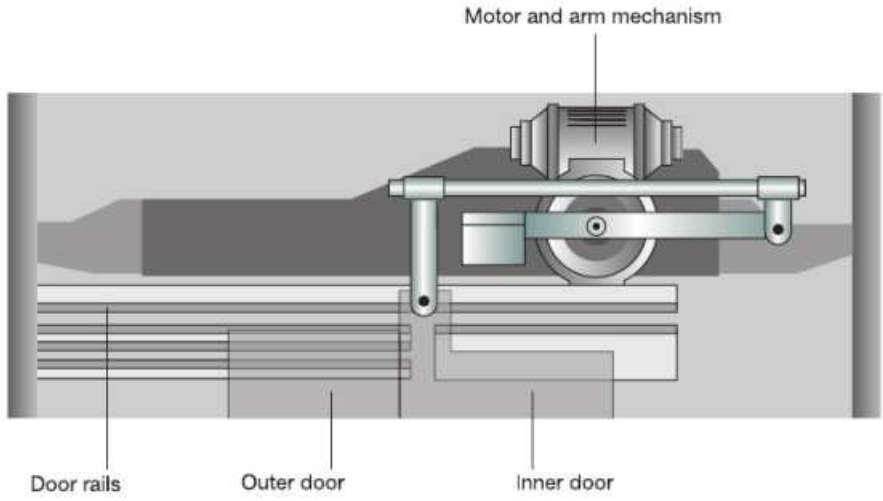




▲ **Figure 2.16** Astronauts on the International Space Station use control software to move a robotic arm into position in order to capture a supply capsule

Control software takes input from one or more sensor(s), makes a decision based on the input value and then outputs something, such as a command, as a result.

7 PREZI IS A TRADEMARK OF PREZI, INC. ALL RIGHTS RESERVED.



▲ **Figure 2.17** Lift doors are opened and closed automatically by control software when a sensor detects that the lift is level with the floor and has stopped

Control software is often used in engineering, vehicles and building control systems. However, a developing group of control applications is in the area of home automation systems, where devices can be monitored and controlled using a smartphone app.



▲ **Figure 2.18** Smartphone apps can be used to control home heating and cooling systems

SKILLS ADAPTIVE LEARNING
INTELLECTUAL INTEREST AND CURIOSITY
COMMUNICATION

ACTIVITY

▼ REMOTE CONTROL

1 Discuss the objects in your home that you think could be connected to the internet in order to be switched on or controlled remotely.

2 Now discuss potential ways in which you could automatically control these connected things in your home based on changes in the environment that are detected by remote sensors or sensors in a smartphone. See Figure 2.19 and the Did you know? feature for some examples.

DID YOU KNOW?

Some smartphone apps can control internet-connected devices in certain situations. For example, if the smartphone calculates that its location is within 10 metres of a garage door, it will send an instruction via the internet to the garage door motors to turn on and open the door.

SKILLS INTELLECTUAL INTEREST AND CURIOSITY
CONTINUOUS LEARNING

ACTIVITY

▼ THE INTERNET OF THINGS

Do some research to find out what is meant by the term, 'the internet of things'.

Control applications are often created by programmers rather than general users and they are not as easy to modify as other application software.

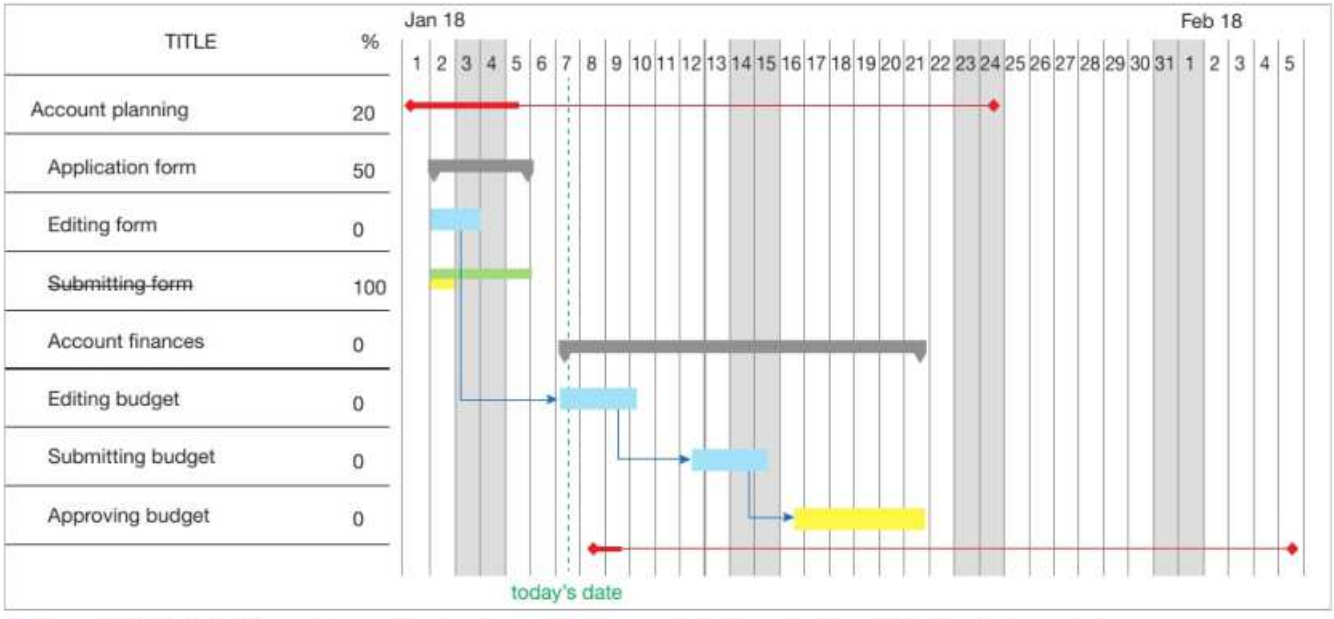
PROJECT MANAGEMENT

GENERAL VOCABULARY

timeline a linear graphical representation of events and the time and order in which they occur
critical path the shortest time a project will take to complete (if no problems or failures are experienced)
milestone a time or date by which a task must be completed

Project management applications are used to help plan and track the individual tasks in a project, so that project managers can make the most efficient use of the available resources. Some tasks cannot be started until a previous task is completed or has been partly completed, so it is important for a project manager to see which tasks are dependent on others. When all of these dependent tasks are combined together into a **timeline**, it is easy for the project manager to see the **critical path**. These applications can also be used to set **milestones**.

Project management applications can be used to allocate tasks to individual people or groups of people. They often provide the option for individuals or groups of people to be allocated to more than one task. Some project management applications also provide tools for tracking the costs and arranging for resources to be delivered on time.



▲ **Figure 2.19** Gantt charts are often used to visualise the timelines and milestones for the individual tasks in a project

COMMUNICATION SOFTWARE

Communication software provides remote access to systems and allows users to contact people using the internet. It can be used to send files and messages as text, images, audio and video.

WEB BROWSERS

Web browsers allow users to view web pages and websites created in web authoring software and hosted on servers that are connected to the internet. These servers may be in another country. Different browsers have different features and compatibility with file types used on the World Wide Web.

Browsers also allow users to access other systems that use the internet but are not on the World Wide Web, such as **file transfer protocol** (FTP) servers.

SUBJECT VOCABULARY

file transfer protocol (FTP) a method of exchanging data and files across a network

EMAIL

Emails can be sent and received by software installed on a computer. This locally installed software acts as a client to an email server (see page 89 for more information about the client-server model). In addition, it often stores or uses a database of contacts, which is sometimes available as another application.

As well as locally installed applications, users can access email through webmail software, which they reach through a web browser. Webmail is an example of a web application that is hosted on a web server and accessed using an internet connection. Some webmail applications allow users to store emails offline for access when they do not have an internet connection.

Email applications often provide features such as labels to help organise emails and **filters**.

SUBJECT VOCABULARY

filter a computer program that only allows certain types of information to pass through it

SOCIAL MEDIA

Social media can be accessed through a web browser or by using locally installed apps.



▲ **Figure 2.20** Social media companies give access to their services through their web pages

Some apps provide access to multiple social media accounts. App developers create apps that people can download and install on their digital devices. TweetDeck is an example of communication software used with social media.



▲ **Figure 2.21** Apps are available for mobile devices as well as PCs

SMS

Short messaging service (SMS) applications are found on mobile phones. They allow users to send up to 160 text characters per message using the mobile phone network. They do not require an internet connection.

MMS

Multimedia messaging service (MMS) applications extend the capabilities of SMS. MMS can deliver more than 160 characters per message and can include video, animations, images and audio. Like SMS, they are sent using the mobile phone network and do not require an internet connection.

INSTANT MESSAGING

Instant messaging applications are very similar to MMS applications, but they require a connection to the internet. They can allow users to see when other users are typing, and users can also prevent others from seeing when they are typing. Messages sent via instant messaging applications can include location data.

SUBJECT VOCABULARY

software licence a legal arrangement that gives a user the right to install and use software

SOFTWARE LICENSING

Sometimes, users require a **software licence** to be able to install and use software on a computer.

There are many types of software licence and the details of software licensing are complicated. To make things easier to understand, you can think about the two types of software that are available:

- free or open-source
- proprietary.

DID YOU KNOW?

Using software without a required licence is called software piracy. Software distributed without the right to do so is called pirate software.

FREE OR OPEN-SOURCE

SUBJECT VOCABULARY

free software software that can be modified or distributed by a user
open-source software software for which the source code is made available to users
source code a collection of instructions that forms a piece of software

HINT

The key difference between the terms 'free' and 'open-source' is that 'free' describes the program and 'open-source' describes the source code.

Free software licences give users the right to study, modify, copy or distribute a program. The user can decide if, and how much, to charge for a copy of the software or any service provided by the software. This means that free software can be made available for a fee or free of charge. 'Free' refers to the user's freedom to charge whatever they want because there are no restrictions on the use of the software.

Open-source software licences make the **source code** available to users so that they can modify how the software works, or distribute the modified or unmodified software.

KEY POINT

In the case of free software, the term 'free' does not refer to whether or not users have to pay for the software. It refers to the fact that the software is free from usage restrictions.

PROPRIETARY

SUBJECT VOCABULARY

proprietary software software that is marketed and distributed by its owner under a brand name
freeware proprietary software provided free of charge to users

Proprietary software is software that is marketed and distributed by its owner under a brand name. The software owner can decide the fee for the software and whether or not the software should be distributed.

Proprietary software can be made available for a fee or free of charge. When no fee is required, the software is called **freeware**. Unlike software distributed under an open-source licence, software with a proprietary licence usually does not make its source code available.

SKILLS ADAPTIVE LEARNING
DECISION MAKING
INTERPRETATION

ACTIVITY

▼ CATEGORISING SOFTWARE

Think about the examples of software types that you have read about in this chapter. Group these examples into the categories 'free or open-source' and 'proprietary'.

SUBJECT VOCABULARY

update (noun) a change or addition to a computer file so that it has the most recent information; (verb) to make a change or addition to a computer file so that it has the most recent information
bugs errors in the program's source code
compatibility ability of one device, system or application to work with other devices, systems and applications
usability the ease with which a device or application can be used

GENERAL VOCABULARY

vulnerability weakness or exposure to harm

SOFTWARE UPDATES

Updates to software are released by software developers for the following reasons:

- fixing security **vulnerabilities** or **bugs**
- increasing **compatibility** with newer operating systems
- improving performance and efficiency
- introducing new features
- improving **usability**.

Software updates are usually made available for download from a server on the internet. Some updates can be scheduled to happen automatically when they are released.

It is important to back up a system and files before updating software in case the update introduces problems, such as removing components that other software needs in order to function.

CHAPTER QUESTIONS

1 State what is meant by the term 'application software'. (1)

SKILLS ADAPTIVE LEARNING

2 State **three** types of utility software. (3)

SKILLS INTERPRETATION

3 Describe the role of the operating system. (2)

SKILLS REASONING

4 Explain why utility software allows the user to schedule backup processes. (2)

SKILLS REASONING

5 Explain the benefit of using compression utilities to archive files. (2)

SKILLS INTERPRETATION

6 Describe how presentation software can help a presenter to give a talk. (2)

SKILLS INTERPRETATION

7 Describe **one** use of control software in a relevant industry. (2)

SKILLS CRITICAL THINKING REASONING COMMUNICATION PRODUCTIVITY

8 Discuss why a building company might use project management software. (8)

SKILLS PROBLEM SOLVING

9 State **three** types of service provided by communication software. (3)

SKILLS REASONING

10 State **two** reasons for updating software. (2)